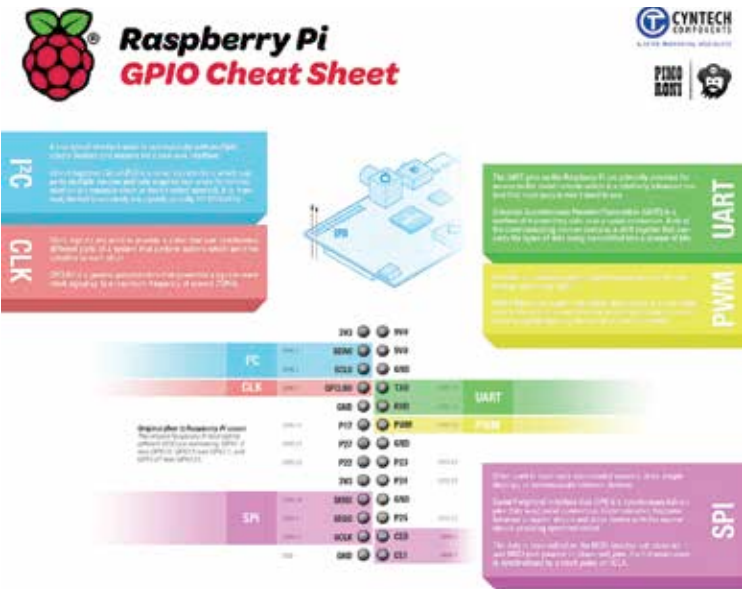




Learning to control the hardware features is a must for hardware driven projects.

These downloaded files need to be run on a functional computer other than the Raspberry Pi. The files need to be extracted to a folder using WinRAR or another similar application. A new file needs to be created in the “source” directory called “main.s”. The “.s” file extension is commonly used for all forms of assembly code, in case of the Raspberry Pi it is meant for the ARM processor. The “main.s” file will be used to carry the code for the new operating system. The “build” directory should be empty, the “main.s” should be the only file in the “source” directory and there should be no other additional files in the extracted folder other than “kernel.ld”, LICENSE and Makefile.

Coding the Compiler

Next you have to open the “main.s” file using a text editor. The assembly code used for the Raspberry Pi and Pi 2 is based on the ARMv6 and ARMv7 processors respectively. The first code in the text editor should be the following:

```
.section .init
.global _start
_start:
```